

Nathan Potraz

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EDUCATION

B.S in Computer Science | California State University, San Marcos • 3.7 GPA • Cum Laude Honors
Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Computer Security, AI, Operating Systems, AGILE/SCRUM.

EXPERIENCE

Hardware & Software Engineer (Founder) | Self-Employed • June 2022 - Present • Oceanside, CA

- Engineered **over 100 custom-tuned** competitive gaming controllers, performing precise circuit-board soldering and firmware optimization to meet professional-grade performance standards.
- Consulted directly with clients** to define technical requirements, translating user performance needs into custom hardware and software specifications.
- Managed full-cycle product development**, including supply chain logistics, hardware debugging, and providing post-deployment technical support.
- Contributed to an open-source motherboard project** by performing board testing, taking electrical measurements, and authoring technical documentation to improve community hardware standards.

PROJECTS

Devlog | devlog.ntpotraz.dev/ • Oct 2025 - Present

- A journaling web application built using **TypeScript React** and a **Golang HTTP server and API**.
- Features simple and fast journal entry **creation, sharing, and editing** utilizing **SQLite hosting** with Turso.
- Integrated Clerk for **secure user authorization**, supporting GitHub and Google SSO as well as email/password.
- Automated deployment with GitHub Actions and Docker, reducing production release time from 15 minutes to under 3 minutes via a CI/CD pipeline targeting Google Cloud Run.

PIC Health & Resource Mobile App | ntpotraz.github.io/PIC-Health-App/ • Aug 2024 - Dec 2024

- Led a 5 developer team** to architect and implement the core UI/UX for a cross-platform community resource hub using **React Native**.
- Served as **Technical Liaison between developers and stakeholders**, translating requirements into actionable sprint goals that increased development velocity by 20%.
- Engineered a **serverless architecture utilizing the Google Calendar API** and Forms, reducing infrastructure costs to \$0 and enabling non-technical clients to manage events seamlessly through Google Calendar.

Japanese-English Translator | github.com/nthpotraz/cs421-translator/ • Mar 2024 - May 2024

- Engineered a C++ natural language processor** using a **Deterministic Finite Automata Scanner** to tokenize and categorize complex linguistic inputs with 100% lexical accuracy.
- Built a Recursive Descent Parser to enforce strict grammar rules, implementing a look-ahead mechanism to handle multi-token syntax structures efficiently.
- Optimized **dictionary lookups to O(1) average time complexity** by implementing custom **Hash Maps**, enabling near-instantaneous translation of large text files compared to standard linear search methods.
- Developed a **translation engine** that programmatically reordered tokens to account for syntactic differences between Japanese and English, ensuring grammatically coherent output.

Gloomy Monsters | github.com/ntpotraz/gloomy-monsters/ • Mar 2020 - Mar 2020

- Created a React-based companion app to **manage complex board game mechanics** and stat tracking for users.
- Designed a **nested state architecture to handle hierarchical data** for multiple active monster groups.
- Integrated a **complex JSON dataset to dynamically populate unit statistics** and ability deck behaviors.
- Applied **conditional visual styling to adjust UI elements** based on specific game rules and unit types.

SKILLS

Backend: Golang, Python, Java, C++, SQL, SQLite, MySQL, PostgreSQL

Frontend: TypeScript, JavaScript, PHP, React, React Native

DevOps/Tools: Linux, Docker, CI/CD, Git, Agile, AWS, GCP